

# Sabatier Carbon Dioxide Methanation Catalytic Kinetics

---

## 1. Reaction Stoichiometry

CO<sub>2</sub> and hydrogen react exothermically:  $\text{CO}_2 + 4 \text{H}_2 \rightarrow \text{CH}_4 + 2 \text{H}_2\text{O}$  ( $\Delta H = -165 \text{ kJ/mol}$ ). Ruthenium-on-alumina catalyst beads operate at 380 degrees Celsius.

## 2. Water Condensation & Electrolysis

Produced water vapor is condensed in finned heat exchangers, and routed to PEM water electrolyzers to regenerate hydrogen and deliver pure oxygen to the outpost.

## 3. Methane Liquefaction Train

Pure methane gas is compressed through a three-stage piston compressor to 35 bar, desiccated through 3A molecular sieves, and liquefied into cryogenic storage tanks.